SINKGRAB · USER REQUIREMENTS

SinkGrab user requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	PR-012 and PR-013 accepted by Amish on 2026-10-25 (SKG-DEC-001 O3)
0.1	2026-10-24	Amish Chadha	First user requirements (kit 1.9.1 upgrade), drawn from SKG-PRB-001 Tables 1 and 2 and the operating environment

SinkGrab user requirements

What well-digging crews, WASH engineers, well committees and fabricators need, written from their side. They are drawn from the users and stakeholders in SKG-PRB-001 (Tables 1 and 2) and its operating environment, and they have not yet been validated with a crew; co-design sessions with the first partner will revise them. Product requirements that meet these needs are in SKG-PRS-001 (`03-requirements.md`) and the links are in the traceability matrix SKG-RTM-001 (`03b-traceability.md`). Method: STANDARDS section 22.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Performance	Dig below the water table	Deepen a flooded well 2 to 3 m below the dry-season water table without pumping it out	High	Well-digging crews; well committees (PRB Table 1)
UR-002	Safety	Nobody in the well	Do all the digging and undercutting from the surface, so nobody works under water or in bad air	High	Well-digging crews; WASH engineers
UR-003	Performance	Keep the lining straight	Sink the caisson rings evenly, see tilt early and correct it	High	Well-digging crews; WASH engineers
UR-004	Deployment and assembly	Fits the wells we dig	Work inside the caisson ring sizes in use, about 0.8 to 1.3 m inside diameter	High	Well-digging crews (PRB operating environment)
UR-005	Performance	Enough soil a day	Lift enough soil each cycle to finish the wet part of a well in a few working days	High	Well-digging crews; well committees
UR-006	Usability	Effort a crew can keep up	Crank, turn and handle with forces a crew of three to five can keep up all day	High	Well-digging crews
UR-007	Safety	Lifting gear we can trust	Never load the tripod past its rating, whatever the crew does; lifting parts proof-tested before use	High	WASH engineers; owners of the HatchSide tripod (PRB Table 2)
UR-008	Logistics	Portable	Carry every piece by one or two people and move the whole kit in a small pickup	Medium	Well-digging crews; WASH engineers
UR-009	Maintenance	Built and mended locally	Build and repair it from drawings with a stick welder, drill and hand tools, from steel the local stockholder sells	High	Local fabricators
UR-010	Cost	Affordable	Cost a programme or a well committee can meet	Medium	Well committees; WASH engineers
UR-011	Environment and durability	Works in our soils	Dig silt, fine sand, laterite, clayey sand and gravel; boulders and rock handled another way	Medium	Well-digging crews (PRB operating environment)
UR-012	Safety	Safe air at the well	Know the shaft air is safe before work, after breaks and whenever someone leans over the shaft	High	Well-digging crews; WASH engineers; safety authorities
UR-013	Regulatory	A method we can specify	A documented, repeatable method with stop rules, so the work can be specified, trained and inspected	Medium	WASH engineers; groundwater departments

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.

UR-011 and UR-012 had no requirement before kit 1.8; PR-013 and PR-012 now meet them (accepted by Amish on 2026-10-25, SKG-DEC-001 O3).